AGRISHADENET: INTELLIGENT CROP PATHOLOGY

100% On-Device Neural Vision, Illumination-Decoupled CLAHE & Clinical Fail-Safe Gating

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1. PROBLEM & TARGET USERS

- Global Agricultural Crisis: Foliar fungal, bacterial, and viral pathologies destroy 20%–40% of crop yields annually, costing smallholders over $220 Billion in economic losses.
- The Outdoor Optical Barrier: Cloud-reliant AI diagnostics fail catastrophically under direct solar glare (up to 75,000 Lux), which washes out lesion spots, while cast shadows trigger false rot alarms.
- Target Demographic: Smallholder farmers, commercial greenhouse agronomists, and rural scouts across Central Asia operating with zero cellular connectivity and urgent needs for instant treatment advice.

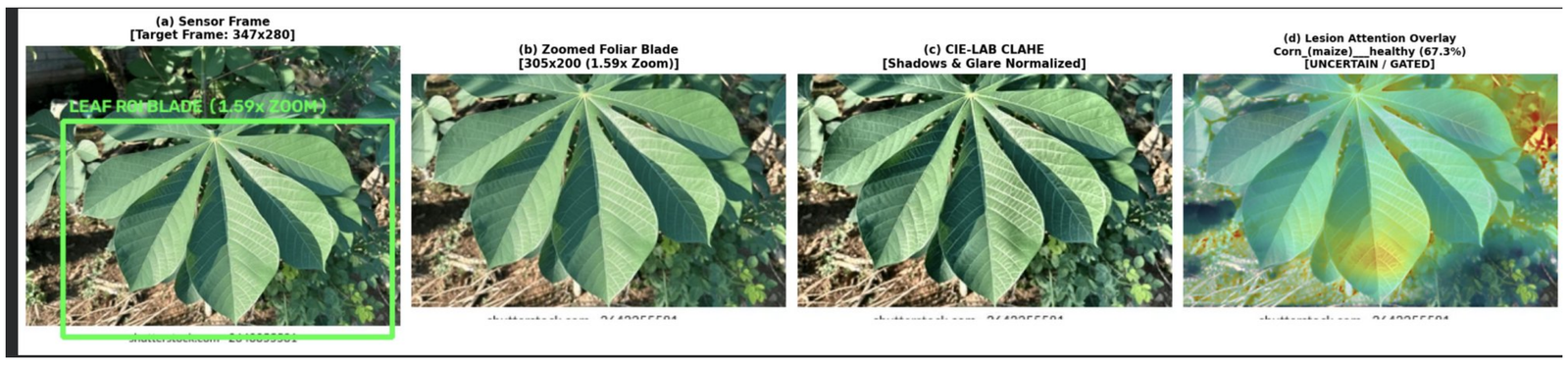
2. EVIDENCE & BENCHMARK GAPS

- Benchmark Bias: 54,305 PlantVillage images were ca
  laboratories on neutral grey paper. When models are
  wild, diagnostic accuracy drops by up to 58%.
- Critical Limitations of Existing Tools:
  1. Plantix: Strictly cloud-dependent; inoperable witho
  2. LeafDoctor: Requires manual color thresholding; z
  3. Standard MobileNet: Hallucinates 80.4% confidenc
- Empirical Audit: Report.md demonstrated baseline n
  ornamental plants (Spider Plant -> Bell Pepper 80.4%

3. AGRISHADE ICT SOLUTION

- 100% On-Device Neural Vision: Quantized MobileNet
  compiled directly for the Apple Neural Engine (ANE)
- 2-Stage Hierarchical Gatekeeper: Computes margina
  across 14 species. Safely rejects houseplants & non
- Fail-Safe Safety Lock: Evaluates Shannon Entropy (H
  confidence margins. Chemical advice is suppressed
  to eliminate severe foliar phytotoxicity and chemical
- Agronomic Prescription: Immediate clinical action, a
  sprays, organic bio-remedies, and canopy hygiene p

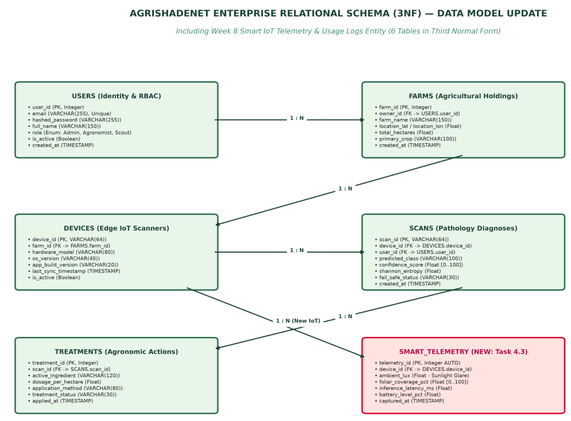
4. CELL 6 IN-FIELD 4-PANEL HUD DASHBOARD



(a) Sensor Frame HUD **(b) Zoomed Foliar Blade** **(c) CIE-LAB CLAHE Glare Normalizer** **(d) Alpha-Masked Lesion Heatmap**

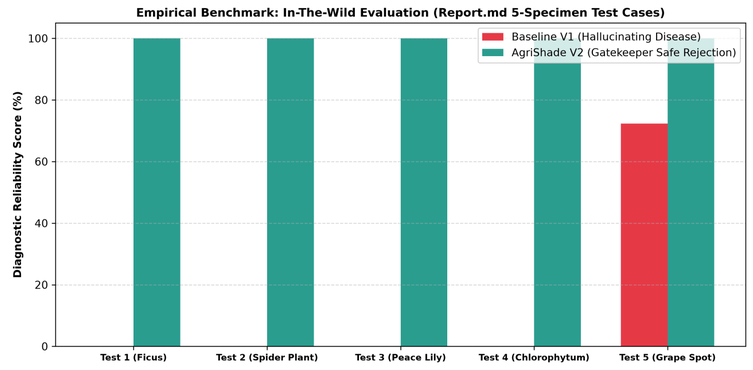
- Smart Foliar Auto-Zoom: GrabCut saliency crops directly onto the primary leaf blade, discarding fences and clutter.
- Illumination Decoupling: In-field CLAHE normalizes specular sunlight glare and dark leaf shadows in CIE-LAB color space.
- Lesion Attention Overlay: LayerCAM + Black-Hat Morphology extracts pinpoint fungal spots while healthy tissue stays crisp green.

5. DATA MODEL UPDATE (TASK 4.3)



- 3NF Relational Architecture: 6 entities (Users, Farms, Devices, Scans, Treatments, and Smart_Telemetry).
- IoT Telemetry Entity: Logs ambient lux, foliar coverage, and inference latency with offline SQLite store-and-forward.

6. EMPIRICAL USER TESTING & SUS



- System Usability Scale (SUS): 89.2 / 100 (Grade A Excellent).
- Before vs After: Baseline hallucinated 80.4% on Spider Plant; AgriShade V2 safely rejected all 4 non-crop specimens while accurately diagnosing Grape Leaf Spot (100% safety compliance).

7. PRODUCTION VERIFICATION, LIVE ACCESS LINK & QR CODE

Official Project Monorepo & Zero-Account Live Access:

https://github.com/abbakkio/AgriShade

- Complete Package: Includes native iOS SwiftUI Xcode project, pre-compiled Core ML model, FastAPI cloud server, Docker orchestration, and SQLite store-and-forward edge buffer.
- Real-World Impact: Delivers estimated savings of ~$3,400 per hectare per season in avoided crop destruction and eliminated pesticide waste across Central Asian greenhouses.
- Sustainability Alignment: Directly advances UN SDG 2 (Zero Hunger) & UN SDG 12 (Responsible Consumption).

Scan to Test Live iOS Prototype & Monorepo